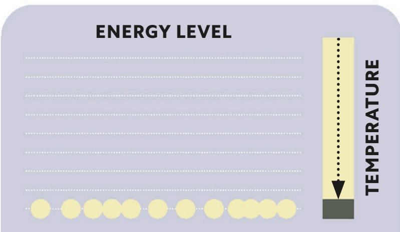
ALTERED STATES

When a large collection of bosons is cooled to a very low temperature, the range of possible energy states available to the particles is greatly diminished, producing a strange form of matter called a Bose-Einstein condensate (BEC). Because the Pauli exclusion principle (see p.69) does not apply to bosons, the particles are not forced to maintain separate states—instead they fall into a shared low-energy state described by a single wave equation.



Critical temperature
Below a certain critical temperature, all the particles fall into the lowest possible energy state, described by a single quantum wave function.

BOSE-EINSTEIN CONDENSATE

As the temperature is reduced toward absolute zero (–459.67°F/ –273.15°C), the wave functions expand still further and merge to form a BEC that behaves as a single giant particle.